6. Confidence Intervals

Dave Goldsman

H. Milton Stewart School of Industrial and Systems Engineering
Georgia Institute of Technology

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Outline

1. Introduction to Confidence Intervals
2. Normal Mean (variance known)
3. Difference of Two Normal Means (variances known)
4. Normal Mean (variance unknown)
5. Difference of Two Normal Means (unknown *equal* variances)
6. Difference of Two Normal Means (variances unknown)
7. Difference of Paired Normal Means (variances unknown)
8. Normal Variance
9. Ratio of Variances of Two Normals
10. Bernoulli Proportion

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Example: $\bar{X}$ is a point estimator for the parameter μ.
A 95% **confidence interval** for μ might look like

$$\mu \in \left[\bar{X} - z_{0.025}\sqrt{\sigma^2/n},\ \ \bar{X} + z_{0.025}\sqrt{\sigma^2/n}\right],$$

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where $z_{0.025}$ is the Nor$(0,1)$'s 0.975 quantile.
This means that μ is in the interval with probability 0.95.

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2	1.90	2.31	2	Yes
3	3.21	3.86	2	No
4	1.75	2.10	2	Yes
5	1.72	2.03	2	Yes
$\vdots$	$\vdots$	$\vdots$	$\vdots$	
10	1.62	1.98	2	No

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The quantity Z is called a **pivot**. It's a "starting point" for us.

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$$\begin{aligned} \mu &\in \bar{X} \pm z_{\alpha/2}\sqrt{\sigma^2/n} \\ &= 278 \pm z_{0.025}(30/5) \\ &= 278 \pm 11.76 \quad (z_{0.025} = 1.96). \end{aligned}$$

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So a 95% CI for μ is $266.24 \le \mu \le 289.76$. $\square$

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Thus, the sample size goes up as...

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Note that we use the $1-\alpha$ quantile (not $1-\alpha/2$).

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so that

$$\bar{X} - \bar{Y} \sim \text{Nor}\left(\mu_x - \mu_y, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right).$$

This implies that

$$Z \equiv \frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim \text{Nor}(0, 1),$$

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$$\begin{aligned}\sum_{i=1}^{n}(X_i - \mu)^2 &= \sum_{i=1}^{n}\left[(X_i - \bar{X}) + (\bar{X} - \mu)\right]^2 \\ &= \sum_{i=1}^{n}(X_i - \bar{X})^2 + 2(\bar{X} - \mu)\sum_{i=1}^{n}(X_i - \bar{X}) + n(\bar{X} - \mu)^2\end{aligned}$$

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\end{aligned}$$

Sketch of Proof: We proved Fact (a) previously. The proof of Fact (b) uses what is known as Cochran's Theorem, which is a bit beyond our scope. In order to derive Fact (c), we first do some algebra to re-write $\sum_{i=1}^n (X_i - \mu)^2$,

$$\begin{aligned}
\sum_{i=1}^n (X_i - \mu)^2 &= \sum_{i=1}^n \left[(X_i - \bar{X}) + (\bar{X} - \mu)\right]^2 \\
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\end{aligned}$$

which can be rewritten as

$$\frac{1}{n-1}\sum_{i=1}^n (X_i - \mu)^2 = S^2 + \frac{n}{n-1}(\bar{X} - \mu)^2. \tag{1}$$

Now note that $X_i \sim \text{Nor}(\mu, \sigma^2)$ and the fact that $[\text{Nor}(0,1)]^2 \sim \chi^2(1)$ together imply

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This process is called **"standardizing and Studentizing."**

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Let's get a 95% CI for the mean residual flame time.

After a little algebra, we obtain

$$\bar{X} \;=\; 9.8525 \quad \text{and} \quad S \;=\; 0.0965.$$

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R Code:

```
x <- data.frame(x=c(
9.85, 9.93, 9.75, 9.77, 9.67,
9.87, 9.67, 9.94, 9.85, 9.75,
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print(confint(lm(x~1,x)))
               2.5 \%   97.5 \%
(Intercept) 9.807357 9.897643
```

Outline

1. Introduction to Confidence Intervals
2. Normal Mean (variance known)
3. Difference of Two Normal Means (variances known)
4. Normal Mean (variance unknown)
5. Difference of Two Normal Means (unknown *equal* variances)
6. Difference of Two Normal Means (variances unknown)
7. Difference of Paired Normal Means (variances unknown)
8. Normal Variance
9. Ratio of Variances of Two Normals
10. Bernoulli Proportion

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Both S_x^2 and S_y^2 are estimators for σ^2 (the common variance).

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After some of the usual algebra, it turns out that

$$\frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(n+m-2).$$

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(Additional technical assumption: All X_i's and Y_j's are jointly normal.)

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Now the problem reduces to the old $\text{Nor}(\mu, \sigma^2)$ case with unknown μ and σ^2.

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Just like before, we get

$$\frac{\bar{D} - \mu_d}{\sqrt{S_d^2/n}} \sim t(n-1).$$

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Clearly, the people are independent, but the times for the same individual to park the two cars may not be independent.

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Bad thing: You'd introduce much more noise into the system by using the "usual" method. This could make the CI much larger.

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The X_i's and Y_i's have the same sample averages as before, but now there's more natural variation, since we're using 10 different people.

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Moral: Use paired-t when you can. $\square$

Outline

1. Introduction to Confidence Intervals
2. Normal Mean (variance known)
3. Difference of Two Normal Means (variances known)
4. Normal Mean (variance unknown)
5. Difference of Two Normal Means (unknown *equal* variances)
6. Difference of Two Normal Means (variances unknown)
7. Difference of Paired Normal Means (variances unknown)
8. Normal Variance
9. Ratio of Variances of Two Normals
10. Bernoulli Proportion

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Thus,

$$\frac{S_y^2/\sigma_y^2}{S_x^2/\sigma_x^2} \;\sim\; \frac{\chi^2(m-1)/(m-1)}{\chi^2(n-1)/(n-1)} \;\sim\; F(m-1,n-1).$$

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$$1-\alpha \quad = \quad P\left(F_{1-\frac{\alpha}{2},m-1,n-1} \leq \frac{S_y^2/\sigma_y^2}{S_x^2/\sigma_x^2} \leq F_{\frac{\alpha}{2},m-1,n-1}\right)$$

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Remark: The CI for σ_x^2/σ_y^2 is proportional to the ratio of the sample variances, S_x^2/S_y^2, and contains no reference to μ_x or μ_y.

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Remark: If you want CIs for σ_y^2/σ_x^2, just flip all of the X's and Y's in the various CIs discussed above.

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$$\frac{\sigma_A^2}{\sigma_B^2} \leq \frac{S_A^2}{S_B^2} F_{\alpha,n_B-1,n_A-1} = \frac{100}{70}(2.11) = 3.01. \quad \square$$

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(If n isn't large, then we'll have to use nasty Binomial tables, which I don't want to deal with here!)

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$$n \;\geq\; z_{\alpha/2}^2/(4\epsilon^2).$$

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So now we only have to take $n = 897$. $\square$

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Coming Up: Hypothesis Testing!